Nexidia Real-Time Grid Admin Guide

Copyright

For assistance, contact your local supplier or nearest NICE Customer Service Center:

EMEA Region (Europe, Middle East, Africa)
Tel: +972-9-775-3800
Fax: +972-9-775-3000
email: support@nice.com

APAC Region (Asia/ Pacific)
Tel: +852-8338-9818
Fax: +852-2802-1800
email: support.apac@nice.com

North America
Tel: 1-800-663-5601
Fax: +201-356-2197
email: na_sales@nice.com

International Headquarters-Israel
Tel: +972-9-775-3100
Fax: +972-9-775-3070
email: info@nice.com

The Americas Region (North, Central, South America)

Israel
Tel: 09-775-3333
Fax: 09-775-3000

France
Tel: +33-(0)1-41-38-5000
Fax: +33-(0)1-41-38-5001

Hong-Kong
Tel: +852-2598-3838
Fax: +852-2802-1800

Tel: 1-800-6423-611 Fax: +720-264-4012 email: support.americas@nice.com	email: support@nice.com
United Kingdom Tel: +44-8707-22-4000 Fax: +44-8707-22-4500	**Germany** Tel: +49-(0)-69-97177-0 Fax: +49-(0)-69-97177-200

NICE invites you to join the NICE User Group (NUG).

Visit the NUG Website at www.niceusergroup.org; and follow the instructions.

All queries, comments, and suggestions are welcome! Please email: nicebooks@nice.com.

For more information about NICE, visit www.nice.com

Install the Application Components

Overview

Nexidia Real-Time Grid (RTG) performs phonetic analysis on telephone calls in real-time. RTG is not a stand-alone application. It is a service-based application framework that provides phonetic analysis capabilities for a client application.

In an Engage environment, Real-Time Grid may forward Engage Business Data fields in call metadata, representing custom data in Engage. Business Data is customized per deployment. Please see the *Engage System Administrator Configuration Guide* for instructions on configuring Engage Business Data.

Install Nexidia Real-Time Grid

Introduction

The Real-Time Grid silent installer is a tool that installs the RTG components. In the directory of the silent installer, there is a batch file called **SilentInstallRTG.bat** that will import the JSON properties file names **silent_install_properties.json** and install any services assigned to the server that the batch file is run on. The files required for the installation are the **silent_install_properties.json**, **keystores**, and **truststores**. To generate these files refer to the *Nexidia Real-Time Grid Silent Installer Guide*.

Install Certificates for Scan SIP/TLS

If Engage media encryption is enabled, Scan must send SIP/TLS to each AIR logger whose calls it will scan. This requires that the customer provide a single signing Trusted Root Authority Certificate, as well as a Server Authentication Certificate, signed by the Trusted Root Authority, for each AIR logger for which calls will be scanned. DER x509 certificates, which typically have a .crt extension, are a format known to work for Scan SIP/TLS.

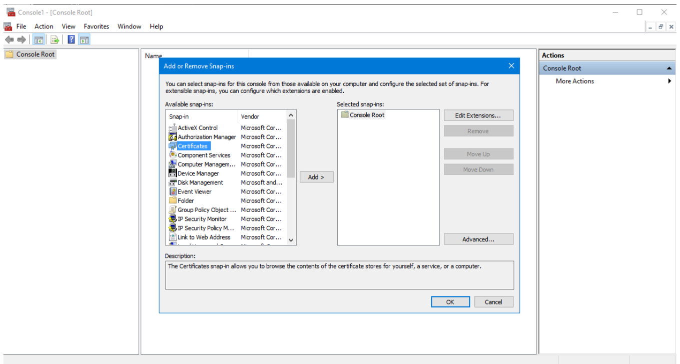
Scan supports ConnectAPI ClientLoginSSO and ClientLoginWithDomain for environments where Engage media encryption is enabled.

For Mattersight TLS setups, to utilize SIPS for communicating to the Mattersight Sip Service (MSS) you need to add the security configuration properties as shown in the [local-properties.xml](#) section and add the trusted root authority to those specified java truststore and keystore files. See [Import the Trusted Root Certificate Authority to the Java Truststore and Keystore](#) below for details.

Verify or Import the Trusted Root Certificate Authority

1. On the *RTG Machine* Run MMC (mmc.exe)

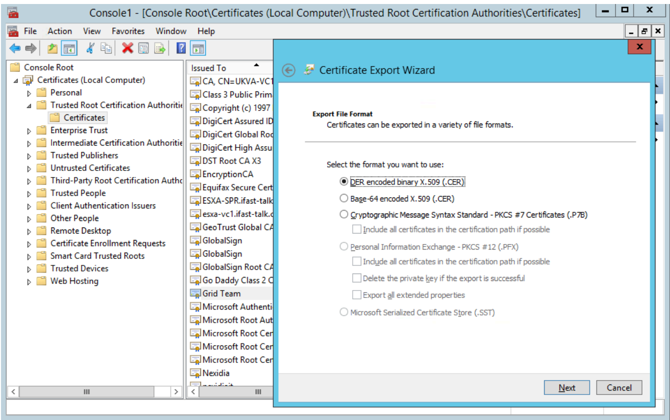
2. Select File→Add/Remove Snap-in…
3. Select Certificates in the left pane, then Add



4. Select Computer account, Next
5. Select Local computer, Finish, then OK
6. In the left pane, expand Certificates→Trusted Root Certification Authorities→Certificates.
7. In the middle pane, verify the customer's signing Trusted Root Certificate appears, indicating it already exists, and does not need to be imported.
8. If the Trusted Root Certificate does not appear in the middle pane, it must be imported. Use the following steps to export the certificate on the NICE Engage Application Server, and then to import it on the *RTG Machine*.

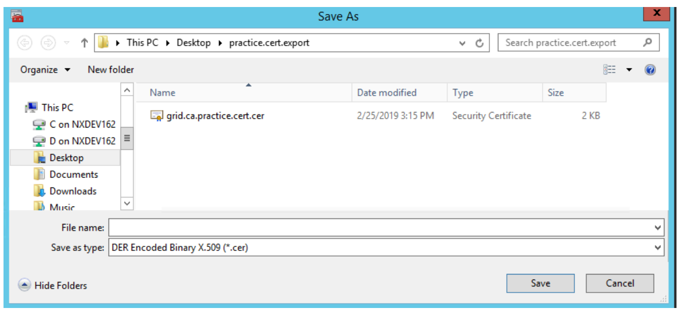
Export the Trusted Root Certificate from the NICE Engage Application Server

1. On the NICE Engage Application Server, use the steps above to run MMC and add the Certificates Snap-in.
2. Navigate to Certificates (Local Computer)→Trusted Root Certification Authorities→Certificates.
3. In the middle pane, right-click the Trusted Root Certification Authority certificate for ConnectAPI/TLS, then All Tasks→Export. The Certificate Export Wizard dialog appears. Click Next.
4. Select DER format, as shown below.



5. Select Next.

6. Use the Browse button to specify a location and name for the exported file. Verify in the Browse window’s Save as type: field that the export will be saved as a DER Encoded Binary X.509 (*.cer), as below. Select Save.



7. Select Next, then Finish. MMC should report "The export was successful."

Import the Trusted Root Certificate on the *RTG Machine*

1. Copy the exported Trusted Root Certificate from above to the *RTG Machine*.

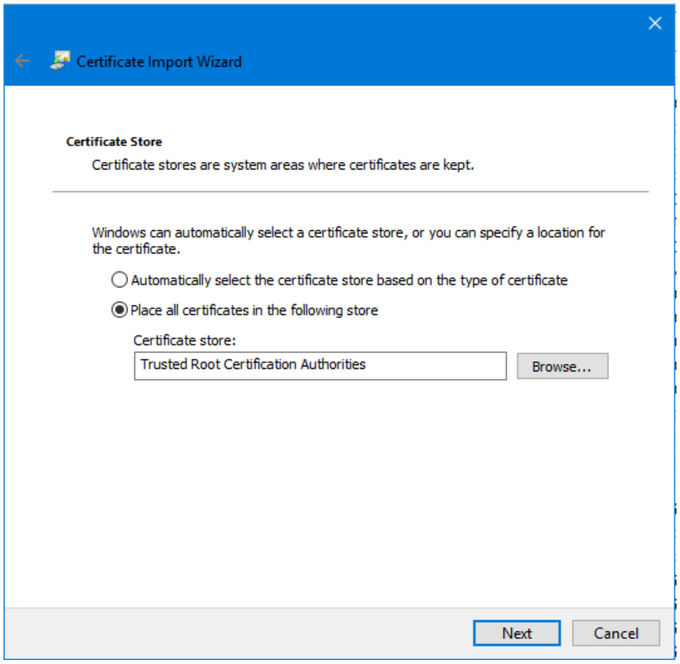
2. Use steps 1-6 above to open MMC on the *RTG Machine*, and navigate to Trusted Root Certification Authorities.

3. In the left pane, expand Certificates→Trusted Root Certification Authorities.

4. Right-click the Certificates folder under Trusted Root Certification Authorities, then select All Tasks→Import→Next→Browse.

5. Navigate to the Trusted Root Certificate file, select Next.

6. Confirm the dialog shows the certificate will be imported to Trusted Root Certification Authorities. If a different certificate store appears, use Browse to select Trusted Root Certification Authorities, then Next, then Finish.



Tip

Scan requires the use of Truststore and Keystore files as described in the "keytool" commands below, as well as importing the certificates using the Microsoft Management Console (MMC). It is not sufficient to only import the certificates using MMC.

Import the Trusted Root Certificate Authority to the Java Truststore and Keystore

The Trusted Root Certificate must be imported into the Java truststore that Scan will use for TLS.

1. Locate the Java truststore that Scan will use.
2. It is possible to use the truststore in <install dir>\jre\lib\security\cacerts, but not recommended, since a subsequent run of the installer will overwrite the cacerts file. It is recommended to copy the cacerts to a permanent location, and then import the Trusted Root Certificate.
3. Use <install dir>\jre\bin\keytool.exe to view the certificates in the truststore.
4. keytool -v -list -keystore c:\path\to\trustore\file
5. If the Trusted Root Certificate from above does not appear, it must be imported. Use the following steps to import the Trusted Root Certificate to the Java truststore. Note that the alias, such as the example below "trustedrootalias," should be all lower-case. Paths specified that include spaces should be enclosed in double-quotes, as below. In this example trusted.root.certificate.crt is an example name for a certificate file.
6. keytool -import -alias trustedrootalias -keystore "c:\path\to\truststore\file" -file "c:\path\to\trusted.root.certificate.crt"
7. keytool will prompt first for the current password, and then for a new password.

8. The default password for cacerts is "changeit." It is recommended that a new password be set when importing the Trusted Root Certificate. Use the [Password Generator] to encrypt the password. Set the property **<entry key="nexidia.realtimegrid.security.truststore.password"></entry>** to the encrypted password.

9. keytool will prompt whether to trust the certificate. Enter "yes."

The Trusted Root Certificate must also be imported to the Java keystore Scan will use. Use steps 3-9 above to import the Trusted Root Certificate into the Java keystore file, substituting the keystore file for the truststore file in steps 4 and 6. The keystore password should be changed. Use the [Password Generator] to encrypt the keystore password and set the property **<entry key="nexidia.realtimegrid.security.keystore.password"></entry>** to the encrypted password.

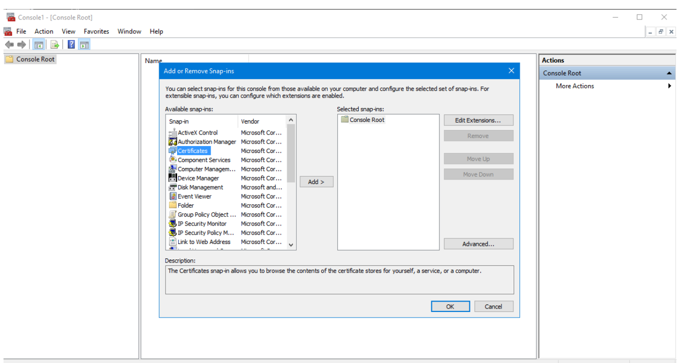
If no keystore file exists, a new keystore will be created when the import command (step 6) is executed.

Import the AIR Logger Server Authentication Certificate

1. On the *RTG Machine* Run MMC (mmc.exe)

2. Select File→Add/Remove Snap-in…

3. Select Certificates in the left pane, then Add

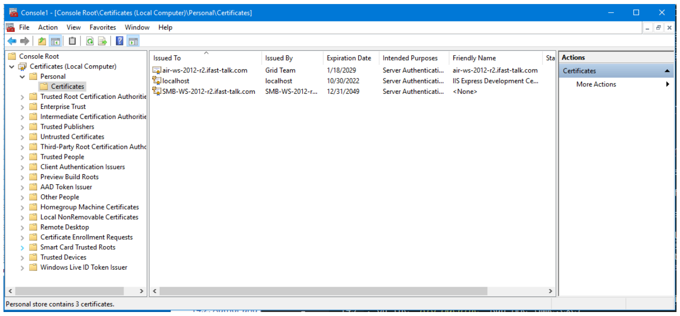


4. Select Computer account, Next

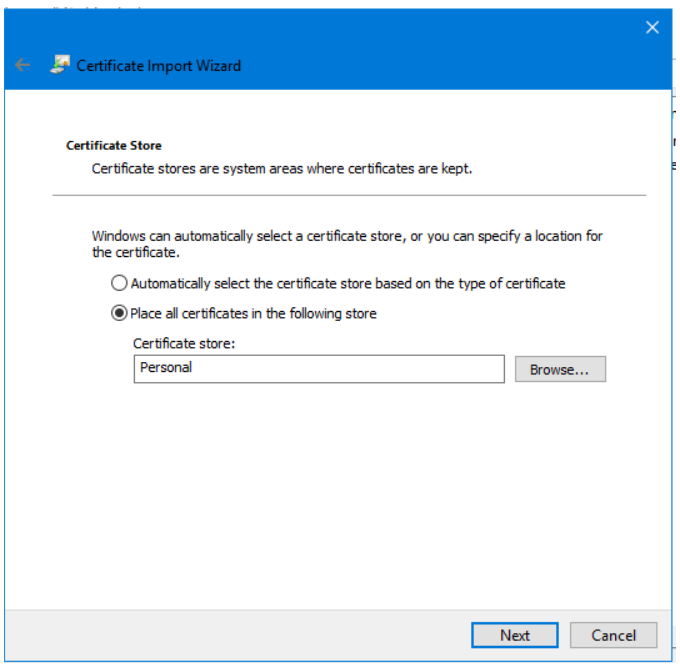
5. Select Local computer, Finish, then OK

6. In the left pane, expand Certificates→Personal→Certificates.

7. In the middle pane, verify the customer's AIR logger Server Authentication certificates have been imported. In the image below, a single AIR logger certificate appears. Note that the key icon is not present, indicating that the private key is not included in the certificate. No private key is necessary for the AIR logger certificates. Each AIR logger certificate should include a Friendly Name.



8. If an AIR logger’s certificate does not appear in the middle pane, it must be imported. Use the following steps to import an AIR logger certificate.
9. In the left pane, expand Certificates→Personal.
10. Right-click the Certificates folder under Personal, then select All Tasks→Import→Next→Browse.
11. Navigate to the AIR logger certificate file, select Next.
12. Confirm the dialog shows the certificate will be imported to Personal. If a different certificate store appears, use Browse to select the Personal store, then Next, then Finish.



Configure Application Components

This chapter provides instructions for configuring the RTG components. The components must be configured after RTG is installed.

Administration

Real-Time Grid includes a Management Console application that allows you to perform several administrative tasks. You will run the Management Console application by selecting:
Start > All Programs > Nexidia > Real-Time Grid > Management Console.

Tip
Enter "get-help" at the console prompt for a summary of available commands.

Management Console Commands

View the existing Scan configuration:

get-config [-o <output>]

Set a new Scan configuration:

set-config -[r][f <file>]

View available commands:

get-help

View installed Workflow(s):

get-workflows

View existing Agents:

get-agent [-o <output>]

Set a new Agent Mapping:

set-agent -[r][f <file>]

View the current metadata that will be appended to call events:

get-append-metadata [-o <output>]

Set new metadata to append to call events or clear existing metadata:

set-append-metadata -[r][f <file>]

View the current air servers:

get-air [-o <output>]

Set new air servers for recording:

set-air -[r][f <file>]

View running RTG services:

get-systemtopology

Exit the Management Console:

quit: exit

Configure Nexidia Real-Time Grid

This topic provides information about configuring the Real-Time grid. You will need to make configuration changes in the following files:

- <install location>\etc\local-properties.xml
- At least one file in <install location>\etc\gateway\telephony-ifc

For more complex deployment scenarios, you may need to make changes to the following files as well:

- <install location>\etc\config\config-service-properties.xml
- <install location>\etc\gateway\gateway-service-properties.xml
- <install location>\etc\scan\scan-service-properties.xml
- <install location>\etc\latency-monitor\latency-monitor-service-properties.xml
- <install location>\etc\telephony-proxy\application.yml

Properties listed as required are present in the relevant files, with default values, after installation. Properties listed as not required may not be present in the relevant files, but they may be copied and pasted from this document.

local-properties.xml

This file contains global properties that would get applied to all the RTG services running on a given machine. Properties can be overridden for a given service by adding the property to the appropriate xxx-service-properties.xml where xxx is the service name (scan, config, gateway, apps).

Property Name	Default	Required	Description
nexidia.realtimegrid.this.bindAddress	127.0.0.1	yes	For a single-se multi-server de

Property Name	Default	Required	Description
			particular RTG
nexidia.realtimegrid.messagebroker.host	127.0.0.1	yes	The server wh required even
nexidia.realtimegrid.messagebroker.port	26010	no	The port that R Message Brok
nexidia.realtimegrid.messagebroker.userName	admin	no	The username Broker.
nexidia.realtimegrid.messagebroker.password		no	The encrypted credentials us
nexidia.realtimegrid.messagebroker.sslEnabled	false	no	When set to tr communicatio `nexidia.re` `nexidia.re` be set and incl
nexidia.realtimegrid.telephony.environment	engage	no	The telephony are: "engage",
nexidia.realtimegrid.connectapi.user		no	The username RTG service u
nexidia.realtimegrid.connectapi.password		no	The encrypted to login to the `nexidia.re` "engage", and
nexidia.realtimegrid.telephony.acdIds		no	A comma delir used to receiv

Property Name	Default	Required	Description
			this should onl `nexidia.re` "mattersight".
nexidia.realtimegrid.telephony.meb.host		no	This field is rec `nexidia.re` "mattersight". Mattersight En
nexidia.realtimegrid.telephony.meb.port	7005	no	The port used (MEB) service `nexidia.re` "mattersight" a
nexidia.realtimegrid.telephony.meb.username		no	The username only be used w `nexidia.re` "mattersight".
nexidia.realtimegrid.telephony.meb.password		no	The encrypted to login to the `nexidia.re` "mattersight".
nexidia.realtimegrid.security.https.enabled	true	no	When set to tr communicatior
nexidia.realtimegrid.security.sips.enabled	true	no	When set to tr (SIPS).
nexidia.realtimegrid.security.client.hostnameVerificationEnabled	true	no	Determines wh server's FQDN certificate retu the DNS and/c

Property Name	Default	Required	Description
nexidia.realtimegrid.security.keystore.path		yes	Path to the jav Contains the e Password Ger
nexidia.realtimegrid.security.keystore.type	jceks	no	Java keystore JCEKS types a
nexidia.realtimegrid.security.keystore.password		yes	The encrypted Generator to tl communicatior **showEncrypt**
nexidia.realtimegrid.security.truststore.path		yes	Path to the Jav Required if `ne` or `nexidia.` true.
nexidia.realtimegrid.security.truststore.type	jks	no	Java truststore PKCS12, JCE
nexidia.realtimegrid.security.truststore.password		no	The encrypted Java truststore Required if `ne` or `nexidia.` true.
nexidia.realtimegird.security.login.method	SSOLogin	yes	Method used t values are Pas environments ClientLoginWit
nexidia.realtimegrid.security.domain.name		no	Name of the c service logs in ClientLoginWit

Property Name	Default	Required	Description
nexidia.realtimegrid.scan.workflow.stream.mode	LOW_LATENCY_FINALIZING	no	The Workflow LOW_LATENC LOW_LATENC
nexidia.realtimegrid.gateway.telephony.protocol	http	no	The protocol u "http" or "https
nexidia.realtimegrid.config.webserver.protocol	http	no	The protocol u "http" or "https nexidia.realtim nexidia.realtim location of the [configuration](#) a
nexidia.realtimegrid.scan.endcall.wait.seconds	45	no	The length of t listen for any r nexidia.realtim The scan serv `nexidia.re` should be set t endcall.wait.se prevent termin exceeded.

Example local-properties.xml

```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE properties SYSTEM "http://java.sun.com/dtd/properties.dtd">
<properties>
  <entry key="nexidia.realtimegrid.messagebroker.host">192.0.2.1</entry>
  <entry key="nexidia.realtimegrid.this.bindAddress">192.0.2.1</entry>
  <entry key="nexidia.realtimegrid.telephony.environment">mattersight</entry>
  <entry key="nexidia.realtimegrid.telephony.meb.host">192.0.2.2</entry>
  <entry key="nexidia.realtimegrid.telephony.meb.username">admin</entry>
  <entry key="nexidia.realtimegrid.telephony.meb.
password">traiA9GhlL6iAmxkrShVrg==</entry>
  <entry key="nexidia.realtimegrid.telephony.acdIds">acd01,acd02,acd03</entry>
</properties>
```

Example agentidmapping.json

The agentidmapping.json file can only be used when DEF server is not available in testing environment. Use curl -X PUT -H "Content-Type: application/vnd.nexidia.realtimegrid+json" http://[config server]:26002/agent -d [file location] to set the agent ids.


```
{
  "agentList": [
    {
      "tisIds": {
        "agentId": "123",
        "hubId": "h456"
      },
      "appIds": {
        "NXRTGAgentId": "n123",
        "NXRTGOsLogin": "o456",
        "NXRTGDomain": "d789"
      }
    },
    {
      "tisIds": {
        "agentId": "456",
        "hubId": "456a"
      },
      "appIds": {
        "NXRTGAgentId": "n123a",
        "NXRTGOsLogin": "o456a",
        "NXRTGDomain": "d789a"
      }
    }
  ]
}
```


apps-properties.xml

The apps-properties file has no user-editable properties. Properties should not be added except as directed by Nexidia technical support personnel.

telephony-ifc server-x-properties files

These files are not relevant in an Engage or Mattersight integration and will be ignored. The files are only being used in a Knoahsoft integration. Each file in the telephony-ifc directory contains connectivity and credential settings for a single Telephony Interface Server. By default, there is one file, server-1-properties.xml, in this location. You may add other server-x-properties.xml files as needed, one for each Telephony Interface Server in your environment. The Gateway Service will read all files in this directory whose names end with properties.xml; therefore, you can rename the server-x-properties.xml file to something more meaningful.

The IP address of the server hosting the Telephony Interface Server.

```
<entry key="nexidia.realtimegrid.telephony.host">127.0.0.1</entry>
```

```
Required: Yes
```

The port on which the Telephony Interface Server is listening.

```
<entry key="nexidia.realtimegrid.telephony.port">9090</entry>
Required: Yes
```

The user name to include in the credentials used to log in to the Telephony Interface Server.

```
<entry key="nexidia.realtimegrid.telephony.username">admin</entry>
Required: Yes
```

The password to include in the credentials used to log in to the Telephony Interface Server.

```
<entry key="nexidia.realtimegrid.telephony.
password">traiA9GhlL6iAmxkrShVrg==</entry>
Required: Yes
```

config-service-properties.xml

The directory where the Config Service should store data such as Workflow models, configurations, and log files. Note that backslashes (\) are not permitted in the path; backslashes must be replaced by either forward slashes (/) or double backslashes (\\).

```
<entry key="nexidia.realtimegrid.config.dataDir">(data location)/config</entry>
Required: Yes
```

The encryption directory that the Config Service should use to store sensitive data. Note that backslashes (\) are not permitted in the path; backslashes must be replaced by either forward slashes (/) or double backslashes (\\). If not set the value of nexidia.realtimegrid.config.dataDir is used.

```
<entry key="nexidia.realtimegrid.config.encryption.dataDir">(data
location)</entry>
Required: No
```

The port that components will use to communicate with the Config Service.

```
<entry key="nexidia.realtimegrid.config.webserver.port">26002</entry>
Required: No
```

scan-service-properties.xml

Directory where the Scan Service should store Workflow models, configurations, and log files. Note that backslashes (\) are not permitted in the path; backslashes must be replaced by either forward slashes (/) or double backslashes (\\).

```
<entry key="nexidia.realtimegrid.scan.dataDir">(data location)/scan</entry>
Required: Yes
```

Maximum number of simultaneous calls that this Scan Node will handle.

```
<entry key="nexidia.realtimegrid.scan.maxActiveCalls">120</entry>
Required: Yes
```

The maximum call duration, in hours, the Scan Service will process before abandoning the call.

```
<entry key="nexidia.realtimegrid.scan.maxCallDurationHr">3.25</entry>
Required: No
```

Network address the Gateway Service should use to communciate with the Scan Service. If this property is not set, the value of nexidia.realtimegrid.this.bindAddress, in local.properties, is used instead.

```
<entry key="nexidia.realtimegrid.scan.webserver.bindAddress"/>
Required: No
```

Port the Gateway Service will use to communicate with the Scan Service.

```
<entry key="nexidia.realtimegrid.scan.webserver.port">26032</entry>
Required: No
```

Network address the Scan Service should use to receive audio data from Telephony Interface Servers. If this property is not set, the value of nexidia.realtimegrid.this.bindAddress, in local.properties, is used instead. Not applicable to Engage integrations.

```
<entry key="nexidia.realtimegrid.scan.telephony.rtp.host"/>
Required: No
```

Lowest port number in the port range the Scan Service will use to receive audio data from the Telephony Interface Servers. Each active call requires four ports.

```
<entry key="nexidia.realtimegrid.scan.telephony.rtp.startingPort">27000</entry>
Required: No
```

Interval at which Enlighten score updates will publish. Default is 10s.

```
<entry key="nexidia.realtimegrid.scan.enlighten.publishing.
frequencySeconds">10</entry>
Required: No
```

Configures whether Scan will publish the transcript for a call. Default is false.

```
<entry key="nexidia.realtimegrid.scan.transcript.publishing.
enabled">false</entry>
Required: No
```

Interval at which Scan will publish transcript messages. Default is 0, which is to publish in real-time.

```
<entry key="nexidia.realtimegrid.scan.transcript.publishing.
frequencySeconds">0</entry>
```

```
Required: No
```

Configures whether Scan will publish the "finalized" transcript for a call. Default is false. Only used when nexidia.realtimegrid.scan.workflow.stream.mode is LOW_LATENCY_FINALIZING.

```
<entry key="nexidia.realtimegrid.scan.transcript.publishing.finalized.
enabled">false</entry>
Required: No
```

Interval at which Scan will publish "finalized" transcript messages. Default is 0, which is to publish in real-time. Only used when nexidia.realtimegrid.scan.workflow.stream.mode is LOW_LATENCY_FINALIZING.

```
<entry key="nexidia.realtimegrid.scan.transcript.publishing.finalized.
frequencySeconds">0</entry>
Required: No
```

Endpoint for the Engage ConnectAPI Login Service

```
<entry key="nexidia.realtimegrid.scan.telephony.connectapiLoginEndpoint">
  https://address.of.nice.app.server/NICEConnectAPI/Login.svc
</entry>
Required: Only if Scan will send SIP/TLS.
```

Transparent NTLM HTTP authentication mode property. Must be allHosts to enable for all hosts. Can be trustedHosts to enable only for hosts that are trusted in Windows Internet settings. By default, the Transparent NTLM HTTP authentication is disabled.

```
<entry key="nexidia.realtimegrid.service.jvm.option">
  -Djdk.http.ntlm.transparentAuth=allHosts
</entry>
Required: Only if Scan will send SIP/TLS and if the ConnectAPI login method is
ClientLoginSSO
```

Interval at which Scan will renew its ConnectAPI login token.

```
<entry key="nexidia.realtimegrid.scan.telephony.
connectapiLoginIntervalSeconds">180</entry>
Required: Only if Scan will send SIP/TLS.
```

Backoff interval, after a failed ConnectAPI login attempt, before Scan will attempt the next login.

```
<entry key="nexidia.realtimegrid.scan.telephony.
connectapiLoginBackoffIntervalSeconds">3</entry>
Required: Only if Scan will send SIP/TLS.
```

Port Scan uses to send SIP requests to Engage AIR loggers. Must be 5061 if Scan will send SIP/TLS.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.uasPort">5064</entry>
Required: No
```

Port Scan uses to receive SIP responses from Engage AIR loggers. 6054 is used only with unencrypted SIP. When SIP/TLS (SIPS) is enabled, this value is ignored and the receiving port is negotiated during the SSL handshake.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.uacPort">6054</entry>
Required: No
```

Protocol Scan uses to send/receive SIP messages to/from Engage AIR loggers. Must be "tls" if Scan will send SIP/TLS (SIPS). Must be "tcp" if `nexidia.realtimegrid.telephony.environment` is set to "mattersight" (should match what is configured in the Mattersight Sip Service (MSS) configuration [udp/tcp]).

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.sipTransport">udp</entry>
Required: No
```

Directory to output JAIN SIP (JSIP) stack related log files.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.sipStackLogDir"></entry>
Required: No (Yes if sipStackLogLevel > 0)
```

Logging level for the JSIP stack. 0=off, 16=DEBUG, 32=TRACE. Do not use any value but "0" in production.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.sipStackLogLevel">0</entry>
Required: No
```

Value of the Sec attribute in a Scan SIP INVITE sent to an AIR logger.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.sipMreqSec">Plain</entry>
Required: No
```

Value of the Encryption attribute in a Scan SIP INVITE sent to an AIR logger. Must be False to receive RTP, True to receive S-RTP. S-RTP is not supported at this time.

```
<entry key="nexidia.realtimegrid.scan.telephony.sip.
sipMsrcEncryption">False</entry>
Required: No
```

latency-monitor-service-properties.xml

The directory where the Transcript Latency Monitor (TLM) Service should store data such as its output CSV and log files. Note that backslashes (\) are not permitted in the path; backslashes must be replaced by either forward slashes (/) or double backslashes (\\).

```
<entry key="nexidia.realtimegrid.latency.monitor.dataDir">(data
location)/latency-monitor</entry>
Required: Yes
```

Location of the CSV file in which TLM records the latency statistics.

```
<entry key="nexidia.realtimegrid.latency.monitor.stats.csv">(data
```

```
location)/latency-monitor/latency.monitor.csv</entry>
Required: Yes
```

Number of threads TLM will use for processing incoming Transcript messages.

```
<entry key="nexidia.realtimegrid.latency.monitor.threadpool.size">5</entry>
Required: No
```

Frequency, in seconds, with which TLM will record a snapshot to the CSV. Default is one hour.

```
<entry key="nexidia.realtimegrid.latency.monitor.snapshot.
frequencyS">3600</entry>
Required: No
```

Name of topic (Transcript) TLM monitors.

```
<entry key="nexidia.realtimegrid.latency.monitor.topic">Nexidia.RTG.
Transcript</entry>
Requred: No
```

Interval, in seconds, between resetting collected latency statistics. Default is to not reset. Note that without reset, once many Transcripts have been received, it will become less likely that outliers will affect the average latency.

```
<entry key="nexidia.realtimegrid.latency.monitor.stats.resetPeriodS">0</entry>
Required: No
```

Maximum number of lines TLM will record in the CSV. Once the maximum is reached, older lines will be removed from the top of the CSV.

```
<entry key="nexidia.realtimegrid.latency.monitor.csv.maxLines">200</entry>
Required: No
```

application.yml

The Telephony Proxy Service configuration file.

Section	Property Name	Default	Requ
server			
	address	127.0.0.1	yes
	port	26004	yes

Section	Property Name	Default	Requ
general			
	data-dir		yes
	encryption-data-dir		no
ss			
	security-enabled	false	no
	https-enabled	false	no
	trust-store		no
	trust-store-type	JKS	no
	key-store		yes
	key-store-type	JCEKS	no

Section	Property Name	Default	Requ
	domain		no
	login-method	SSOLogin	yes
	hostname-verification-enabled	false	no
	sips-enabled	false	no
passwords			
capis-login			
	destination	https://127.0.0.1/NICEConnectAPI/Login.svc/Rest/	no
	threadpool-size	1	no
	read-timeout	5	no

Section	Property Name	Default	Requ
	connect-timeout	30	no
	backoff-interval	10	no
	login-refresh-interval	300	no
engage			
	sleep-rate-seconds	600	no
	air-whitelist		no
capis-getuserinfo			
	destination	https://127.0.0.1/NICEConnectAPI/UserAdministration.svc/rest/GetUserInfo	no
	num-agents-to-send	100	no
capis-subscribe-			

Section	Property Name	Default	Requ
agent			
	connect-and-subscribe-type	osLoginSingle	no
	num-ids-for-bulk-send	50	no
	sleep-rate-seconds	180	no
	message-timeout-seconds	30	no
	events-to-subscribe	CLS_START_RECORD_EVENT,CLS_END_CALL_EVENT,CLS_END_RECORD_EVENT	no
capis-cti			
	destination	wss://127.0.0.1/NICEConnectAPI/Inter...nnectByOsLogin	no
	agent-type	Extension	no

Section	Property Name	Default	Requ
	audio-codec	G.711u	no
	max-message-size	100000	no
	event-thread-pool-size	5	no
	call-timeout-minutes	150	no
mattersight			
	meb-host	127.0.0.1	no
	meb-port	7005	no
	username		no
	receiving-host	127.0.0.1	no

Section	Property Name	Default	Requ
	receiving-port	26011	no
	context-path	/telephony	no
	acd-ids		no

Example application.yml

```
server:
  address: 127.0.0.1
  port: 8080

ss:
  trust-store: C:\rtg\certs\cacaert
  trust-store-type: JKS
  key-store: C:\rtg\certs\scan.keystore
  key-store-type: JCEKS
  domain: COMPANY-DOMAIN

passwords:
  nexidia.realtimegrid.security.keystore.password: traiA9GhlL6iAmxkrShVrg==
  nexidia.realtimegrid.security.truststore.password: traiA9GhlL6iAmxkrShVrg==

capis-login:
  destination: https://nice-app-server/NICEConnectAPI/Login.svc/Rest/
  threadpool-size: 1
  read-timeout: 50000
  connect-timeout: 50000
  backoff-interval: 60000
  login-refresh-interval: 300000

capis-cti:
  destination: wss://nice-app-server/NICEConnectAPI/InteropDuplex.svc/
websocket/SubscribeAndConnectByOsLogin
  agent-type: Extension
  audio-codec: G.711u
  max-message-size: 100000
```

```
engage:
  sleep-rate-seconds: 60
  air-whitelist: air_server_name1,air_server_name2

capis-getuserinfo:
  destination: https://nice-app-server/NICEConnectAPI/UserAdministration.svc/
rest/GetUserInfo

capis-subscribe-agent:
  connect-and-subscribe-type: osLoginSingle
  message-timeout-seconds: 30
  sleep-rate-seconds: 180
  events-to-subscribe: CLS_START_RECORD_EVENT,CLS_END_CALL_EVENT,CLS_END_
RECORD_EVENT

mattersight:
  meb-host: 10.1.1.20
  meb-port: 7005
  username: admin
  receiving-host: 10.1.1.20
  receiving-port: 26011
  context-path: /telephonyproxy
  acd-ids: acd_123
```

Note **What's next?** After RTG has been configured, use Control Panel→Administrative Tools→Services to start the services. See Start the Services.

Start the Services

After installing and configuring the Real-Time Grid application, you will need to start the services:

1. Go to Services in the Server Manager.
2. Start the following services. Depending on the nodes selected during installation, some services may not appear:
    - NexidiaRtgConfigService
    - NexidiaRtgGatewayService
    - NexidiaRtgTelephonyService
    - NexidiaRtgScanService

Note **What's next?** After RTG services have been started, it is required to install at least one Workflow ZIP Model. See Install a Workflow.

Install a Workflow

RTG requires a Workflow ZIP Model, in order to use the Real-Time Grid applications. Use the Management Console included in the Real-Time Grid installation "bin" directory to add a Workflow.

To add a Workflow ZIP Model:

1. Run Management Console from the start menu. You will run the Management Console application by selecting **Start > All Programs > Nexidia > Real-Time Grid > Management Console**.
2. At the command prompt, enter:

> add-workflow -f <path/to/workflow/zip/model>

For example:

> add-workflow -f "C:\Program Files (x86)\Nexidia\Workflows\NX_RTG_realtime_NorthAmericanEnglish_WB10.6_v5.nxzm.zip"

Tip See Administration for more information about the Management Console.

Password Generator

InstallerDialogBannerImage

Copyright

PROPRIETARY AND CONFIDENTIAL INFORMATION

Information herein is proprietary information and trade secrets of NICE Ltd. and/or its affiliated companies (Affiliates). This document and the information herein is the exclusive property of NICE and its Affiliates and shall not be disclosed, in whole or in part, to any third party or utilized for any purpose other than the express purpose for which it has been provided.

IMPORTANT NOTICE

Subject always to any existing terms and conditions agreed between you and NICE or any Affiliate with respect to the products which are the subject matter of this document, neither NICE nor any of its Affiliates shall bear any responsibility

For assistance, contact your local supplier or nearest NICE Customer Service Center:

EMEA Region (Europe, Middle East, Africa)
Tel: +972-9-775-3800
Fax: +972-9-775-3000
email: support@nice.com

APAC Region (Asia/ Pacific)
Tel: +852-8338-9818
Fax: +852-2802-1800
email: support.apac@nice.com

North America
Tel: 1-800-663-5601
Fax: +201-356-2197
email: na_sales@nice.com

International Headquarters-Israel
Tel: +972-9-775-3100
Fax: +972-9-775-3070
email: info@nice.com

The Americas Region (North, Central, South America)
Tel: 1-800-6423-611
Fax: +720-264-4012
email: support.americas@nice.com

Israel
Tel: 09-775-3333
Fax: 09-775-3000
email: support@nice.com

France
Tel: +33-(0)1-41-38-5000
Fax: +33-(0)1-41-38-5001

Hong-Kong
Tel: +852-2598-3838
Fax: +852-2802-1800

United Kingdom
Tel: +44-8707-22-4000
Fax: +44-8707-22-4500

Germany
Tel: +49-(0)-69-97177-0
Fax: +49-(0)-69-97177-200

NICE invites you to join the NICE User Group (NUG).

Visit the NUG Website at www.niceusergroup.org; and follow the instructions.

All queries, comments, and suggestions are welcome! Please email: nicebooks@nice.com.

For more information about NICE, visit www.nice.com

A command line tool used to generate an encryption key in the keystore file and encrypt passwords stored in the properties files.

1. Navigate to the RTG installation bin directory. The default location is `C:\Program Files\Nexidia\Real-Time Grid 12.5\bin`. Open a command prompt in the directory.

2. Run the command password-generator.cmd with the following parameters:

```
-keystoreFilename <keystore_filename>
-keystorePassphrase <keystore_password>
-keystoreType <keystore_type>
-generateEncryptionKey
-showEncryptedPassphrase
-passwords <password_list>
```

<keystore_filename> is the path of the keystore file where the encryption key exists. The file must exist. Required

<keystore_password> is the password of the keystore file. Required

<keystore_type> is the keystore type. The choices are JCEKS (Default) or PKCS12. Optional

-generateEncryptionKey generates the encryption key and stores it in the file defined by the *<keystore_filename>* value in the **-keystoreFilename** parameter. Optional

-showEncryptedPassphrase outputs the keystore password defined by the *<keystore_password>* value in the *-keystorePassphrase parameter. Optional

<password_list> comma delimited list of passwords or strings to encode. Optional

3. The output will show each password and its encrypted value.

The following example would encode a SQL connection string:

password-generator.cmd -keystoreFilename ./example.keystore -keystorePassphrase [unencrypted keystore password] -keystoreType JCEKS -generateEncryptionKey -showEncryptedPassphrase -passwords jdbc:sqlserver://example.sql.server.addre...xamplepassword

Passphrase encrypted: [encrypted keystore password] Passwords Encrypted: jdbc:sqlserver://example.sql.server.addre...xamplepassword: [encrypted SQL connection string]